

Efficient Boxing Punch Classification: Fine-Grained Skeleton-based Recognition Made Light

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ABSTRACT

Sports analytics is a field of study that utilizes camera and sensor data to monitor the athlete's performance and health to optimize the player's strategy and increase the success rate. Coaches rely on analytics to scout opponents and optimize play calls in gameplay. With the advancement in artificial intelligence, accessible and in-depth data collection has been enabled. The well-grounded technique for performance evaluation in sports analytics is Human Pose Estimation (HPE). Our focus is on real-time action recognition in combat sports like boxing. Existing state-of-the-art deep learning models are heavily parameterized, so can't be used in real-time in any low-end devices. Apart from this, fine-grained classification in highly dynamic activities in sports are typically performed using sensors only. Our proposed Machine Learning based pipeline provides real-time fine-grained solution for 14 boxing punch types of classification using RGB video only. Our approach includes the implementation of three novel and generalized motion dynamics features that encode velocity as well as acceleration of the pose sequences., 1) Unified-Axis Angular Encoding (UAE), 2) 2D Motion Dynamics Descriptors (2DMDD), 3) Fifth-order Angular Encoding (FAE). We employed classical machine learning algorithms I.e., Support Vector Machine (SVM), Random Forest (RF), and K Nearest Neighbours (KNN) to make a lightweight model and test it on YouTube videos. The average accuracies of pipeline using the proposed features are found to be **55%**, **92%** and **84%** for UAE, 2DMDD, and FAE respectively. Using KNN, we have achieved **99%** accuracy on 10-fold cross-validation by using FAE features.

Keywords: Skeleton-based Action Recognition, Human Pose Estimation, Computer Vision, Sports Analytics

1. INTRODUCTION

Human Action Recognition (HAR) is a versatile AI segment with many uses in healthcare, fitness tracking, security, sports analytics, human-computer interaction, video surveillance, etc. Technology-aided sports analysis is a rapidly growing field in personal and commercial space. Well-structured statistics of players help in training and judging as they quantify a player's potential and the value he can add to the team. Sports analytics not only judges a player but also the health and safety of players. Especially in high-intensity sports such as martial arts, boxing, etc., players are at high risk of attaining injuries.

In combat sports, performance is judged based on the players' motion, i.e., trajectory. The numerical representation of players' trajectories is hard to interpret, even for experienced people. Recent advancements in Virtual and Augmented Reality[1, 2] have made it feasible to represent natural objects in virtual spaces. Hence, Boxing Analytics using VR/AR technologies can provide brisk interpretability.

Automatic recognition of punches in boxing is crucial before quantifying a player's performance. Most State-of-the-Art boxing punch classification methods use IMU sensor data to classify the punches. Some studies use visual data [3, 5, 6, 7, 8, 9] instead, but either need costly depth measuring sensors or could be faster due to naive trajectory feature extraction. Also, those models use heavy Deep Learning classifiers, which require vast amounts of data, extensive for portable devices. Considering a player with limited resources, i.e., only a smartphone, there is a need for Lightweight models such as classical ML-based algorithms. For this purpose, velocity and acceleration encoded feature are derived from the pose sequences using the angle between joint pairs.

Our main contributions are: 1) Novel sports-specific fine-grained classification dataset, 2) novel feature extraction methodology, 3) vision-only based fine-grained boxing punch classification.

2. RELATED WORK

Human Action Recognition is a vibrant and rapidly evolving field with diverse applications, from enhancing human-computer interactions to bolstering security systems. In this section, we delve into two pivotal approaches: vision-based and sensor-based methods, each representing a substantial body of research and innovation. In vision-based classification, input or data is intensity values at different time instants; the pipeline consists of computer vision principles to extract features. On the other hand, we could use on-body sensors attached to prominent action-contributing joints to get kinematic and orientation data, which acts as the basis for further abstract features in the pipeline. Further in this section, we explained both methods.

2.1. Vision-Based HAR

Visual data for HAR can be of two types; they are 2D and 3D. 2D imagery is straightforward to capture and to work. However, they have an intrinsic self-occlusion problem. 3D imagery based HAR can be very accurate as it can capture the causal pose during action despite occlusions. However, it requires very sophisticated sensors to capture high-speed motion.

The Primitive Method for Human Action Recognition employed RGB images in conjunction with optical flow, coupled with Deep Networks^[9, 10, 11] to recognize actions. Optical flow provided crucial information about trajectory and temporal aspects. Several studies such as Krupa et al.^[6] and Hemayed et al.^[7], harnessed trajectory descriptors from optical flow to classify general human actions. However, these approaches demanded substantial computational resources, limiting their real-time applicability. On the other hand, Zhou et al.^[5] adopted dense-trajectory descriptors, correlation, and weighted trajectory encoding as features extracted from 2D visual data, achieving accurate human action classification.

Pose-based Human Action Recognition (HAR)^[8, 12, 13] has emerged as a more promising approach, focusing on human landmarks as perceptible cues for action understanding. Lin et al.^[14] introduced a time-series modeling of HAR by integrating Pose Estimation with Long Short-Term Memory (LSTM) networks.

Holanda et al.^[3] proposed a rapid human action recognition method using visual question-answering techniques, processing a 2D image frame in a mere 190 milliseconds. Their approach involved an ensemble of classifiers, including Bayesian, Multinomial, Complement, and Gaussian classifiers. These classifiers were applied to same-side and cross-side Euclidean distances between selected joints, as well as knee, elbow, and shoulder joint angles. Additionally, relative KeyPoint coordinates relative to the nose, assumed to be a static point during the action, were incorporated as features. It's worth noting that this feature set, while comprehensive, may benefit from further optimization, especially in the context of boxing classification. S.Kasiri et al.^[4] classified boxing punches using overhead depth imagery captured using visual and depth sensors (they have depth information and only overhead view costly sensors). They use on the upper body of the player for punch classification, 3D features such as the direction of forearm movement, the elbow angle features along with transition-invariant trajectory features, Fourier magnitude, 1st, 2nd, 3rd order central tendencies of hand trajectory are fed to multi-class Support Vector Machine (SVM) classifier and Random Forest in sets containing various combinations of the features. Vision-based Action Recognition is between a tradeoff of fast, cheap 2D data with occlusion problems and costly, sophisticated 3D. With most publicly available data being 2D and Pose-based HAR being better than optical flow-based due to the reduced point of interest, we will work with 2D pose data in our method. Due to high-speed action and self-occlusion problems, it is intrinsically challenging to classify boxing punches using 2D data, but this can be done with better feature extraction.

2.2. Sensor Based HAR

In recent years, the development of sensor-based systems for boxing punch classification has garnered significant attention within sports technology. This surge in interest can be attributed to its capability to record high-precision kinematic data at high temporal resolution, giving very accurate 3D poses. Inertial measurement Unit (IMU) which comprises an accelerometer and gyroscope, is commonly used for HAR. IMUs are placed at human body landmarks of interest; in the case of boxing, it's over left and right shoulders, elbows, wrists, hips, and knees.

Sensor data can capture subtle variations in the action; hence, it can discriminate better than vision-based methods. Vaidyanathan et al.^[15] was able to classify 18 boxing punches and 24 taekwondo kicks using Dynamic Time Warping and CNNs with IMU sensor data. CNNs are widely used to classify time-series data^[16, 17]. Debayle et al.^[17] uses recurrent maps to convert 1D time-series data to 2D texture maps, which are further processed using CNN and classified using Support Vector Machine (SVM). Potentially, many wearable devices are used for personal physiology tracking, which made the

creation of large datasets feasible, such as CAPTURE-24^[18], HHAR^[19], MMact^[20], etc. A lot of research has happened regarding sports analytics using sensor data than vision data.

Many studies, such as the work by Ottersten et al.^[8], have approached boxing punch classification problem using human pose-based trajectories, but we observed that in case of boxing punch classification we can extract better features, which captures causality of the punch that can improve the classifier performance with lesser number of features. Even though sensor data is very accurate in predicting poses, they are expensive and may get damaged during combat sports. Sensors are not allowed during gameplay; vision-based methods are the only affordable and manageable alternative methods. Therefore, our approach uses only 2D video-based poses to classify the various types of boxing punches in real-time.

3. DATASET CREATION

3.1. Dataset Creation Module Overview

In this section, the dataset creation procedure is explained in detail. We recorded various types boxing punches of elite boxers (including male and female) from front and side-view. We employed a pose estimation module to estimate the pixel coordinates of body joints. Given a video of a single player, this dataset creation module processes it and give corresponding body joint pixel coordinates in array format of dimension $F \times W \times 12 \times 2$, F = number of punch instance in an input video, W = number of frames in a datapoint, in each datapoint consist of 2D pixel coordinates of 12 joints, they are left & right shoulders, elbows, wrists, hips, knees and ankles.

3.2. General Pipeline

The first step of the process is to estimate poses in each frame of 60 FPS video. There are many standard methods to estimate poses such as AlphaPose^[14], OpenPose^[15], DeepPose^[16] etc. Many studies^[24, 25, 26] showed that AlphaPose can avoid problems in case of multi-person occlusions as it follows top-down approach and is comparatively fast with enough accuracy, hence we used Alpha Pose for pose estimation. In the case of multi-persons appearing in a frame, we used a bounding box confidence score as a metric to choose the person of interest. After pose estimation, all pixel data is being converted to time-series data, now we move to the second step of the process i.e., Punch Instance(P.I) Detection. In Punch Detection, we first make total frame data into 40 frame batches, study conducted by Thiel et al.^[27] showed that the average time of punch from initiation to completion is around 0.35 seconds for an average human, which can be captured by 21 frames in a 60 FPS video, hence we used 40 frame windowing with 30% overlapping such in worst case it will capture at least one frame in a window. In any kind of punch in boxing at peak instance the elbow will be much closer to corresponding shoulder, we used this observation to detect the peak punch instance. Thiel et al.^[27] also showed that peak instance happens in 0.025 seconds which is 2-3 frames in 60 FPS video, we divided time-series elbow height into 3 frame bin and measured local minimum within 5 bin (15 frames) neighborhood to detect punch instance. $P.I = (3 \times N_{bin} - 1) + S$, $P.I$ is punch instance, N_{bin} local minima bin number, S is frame number of starting frame in a batch. We included no punch instance, considering periodic punching in the captured dataset No Punch happens between two punch instances hence $P.INo = (P.I_n + P.I_{n+1})/2$, where $P.INo$ is no punch instance frame number $P.I_i$ is i th punch instance.

3.3. Classifier Based Dataset

In our methodology, we proposed two types of classifiers: 1. Rule-Based Classifier, 2. Machine Learning (ML) Classifiers. Rule-Based Classifiers classify each frame and works with only side-view whereas ML Classifiers use sequence of frames data and works on front and side view, hence we created two datasets for each method. In Rule-Based Classifier dataset we manually separated all side-view videos for the input and this dataset has a dimension $F \times 1 \times 12 \times 2$ i.e., 1 frame per datapoint. We only consider P.I's frame data. In ML Classifiers, we know that a complete punch can be captured in 20 frames so, we considered 20 frames around the P.I frame. Hence, second dataset has dimensions of $F \times 20 \times 12 \times 2$. Figure 1 describes the whole dataset creation pipeline and separate branches for Rule-Based Classifier and ML Based classifier. Each video will give two files; one that contains punch data labeled with video annotation, other contains no punch instances. But we used only a few no punch data files to keep the dataset unbiased. Table 1 and Table 2 contain information related to dataset distribution of Rule-Based Classifier and ML based classifier respectively.

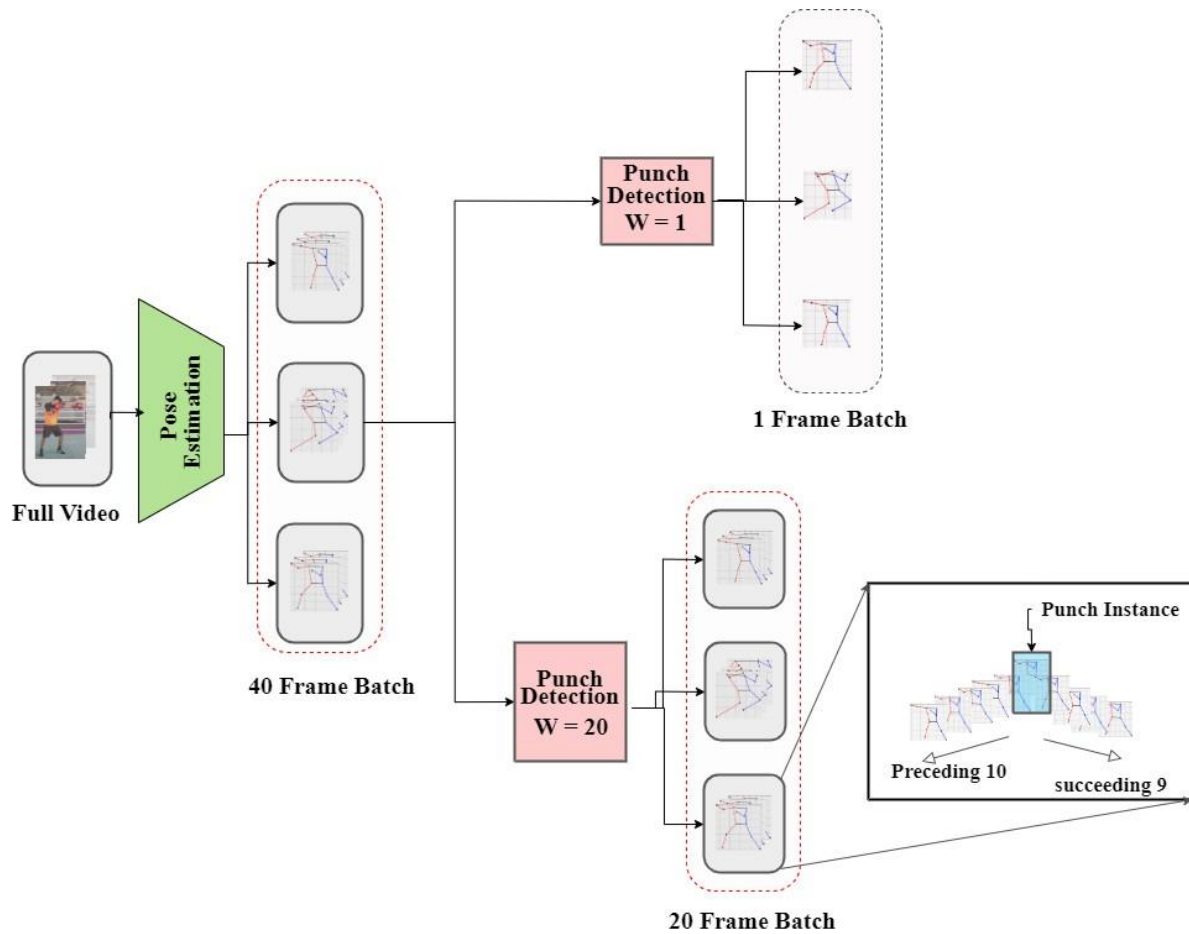


Figure 1. Dataset Creation Pipeline for Rule-Based and Machine Learning Based Classifiers

Table 1. Rule-Based Classifier dataset distribution. 7 Punch Types captured in side-view perspective

S.No	Punch Type	Number of Datapoints
1	Lead Uppercut	60
2	Rear Uppercut	81
3	Lead Hook	45
4	Rear Hook	60
5	Jab	74
6	Cross	59
7	No punch	55

Table 2. Machine Learning Classifier Dataset with 14 punch types from data captured from both front and side-view

S.No	Class	Punch Type	Number of Datapoints
1	0	Lead Jab to head (Long Range)	64
2	1	Rear Jab to head (Long Range)	61
3	2	Lead Jab to body (Long Range)	58
4	3	Rear Jab to body (Long Range)	55
5	4	Lead Hook to head (Long Range)	76
6	5	Rear Hook to head (Long Range)	53
7	6	Lead Hook to head (Mid-Range)	74
8	7	Rear Hook to head (Mid-Range)	63
9	8	Lead steep Hook to head (Mid-Range)	66
10	9	Rear steep Hook to head (Mid-Range)	57
11	10	Lead Hook to body (Mid-Range)	46
12	11	Rear Hook to body (Mid-Range)	50
13	12	Lead Uppercut to head (Mid-Range)	68
14	13	Rear Uppercut to head (Mid-Range)	105

4. METHODOLOGY

Boxing punches are categorized into three primary classes: Uppercut, Straight, and Hook, with further variations based on Stance and Hand-in-action, resulting in Lead and Rear versions. Lead Straight Punch is referred to as Jab, and Rear Hook is simply Hook. The key distinction among these punches is their trajectory: Uppercut moves upward, Hook follows a parallel path to the body, and Straight Punch moves perpendicular to the body plane. We've added a 'No Punch' class, making a total of 7 punch types. These six punch types can also be categorized based on the target (Head or Body), expanding the classes to 14. To distinguish target-based punches, we use machine learning for 14-punch classification, relying on trajectory information and a complex classifier to map pose-trajectory features to classes.

4.1. Rule-Based Classifier

As mentioned before, the causal features of boxing punch trajectories are explained in this section. The Rule-Based Classifier pipeline is visualized in Figure 2. A note that we used causal features to classify 7 types of punches but we limited ourselves to a side view due to self-occlusion issues in the case of a front view, but most of the real world videos are captures such they will have enough part of side-view, this can be seen in Results section where we used our Classifier on YouTube boxing video.

Firstly, Hand Detection plays a pivotal role in determining whether a punch is happening or not, and if it is happening, then which is the active hand during the punch i.e., left or right. We used the relative heights of the right ($l_{es}(r)$) and left ($l_{es}(l)$) elbows from their corresponding shoulders; for all major types of punches, the relative height of the elbow of the hand in action will be less than others. We use three-tenths of the maximum wrist-shoulder length as the threshold. If the

difference between the wrist-shoulder heights of the left and right arm is less than the threshold mentioned, we classify it as No Punch, in case of No Punch we stop further procedures and move to the next frame. Otherwise, we consider the hand with minimum elbow height and send data for further modules.

$$Hand = \{ \underset{\text{argmin}}{\text{min}}(l_{es}(r), l_{es}(l)), |l_{es}(r) - l_{es}(l)| > \frac{3}{10} \max(l_{es}(r), l_{es}(l)) \text{ No Punch}, |l_{es}(r) - l_{es}(l)| \leq \frac{3}{10} \max(l_{es}(r), l_{es}(l)) \} \quad (1)$$

Secondly, Stance Analysis is a crucial differentiator between orthodox and southpaw positions. We derive two essential vectors: the facing vector (V_f), starting from the midpoint of the shoulder to the midpoint of the wrist, and the stance vector (V_s), from the right ankle to the left ankle. The dot product of these vectors informs us whether the stance is orthodox or southpaw; in the case of the orthodox dot product, it is positive and negative in the case of the southpaw.

$$Stance = \{ Orthodox, < V_f, V_s > \geq 0 \text{ Southpaw}, < V_f, V_s > < 0 \} \quad (2)$$

Thirdly, our Major Punch Detection Rule Tree comprises two hierarchical rules, factoring in pose information and the identified active hand as inputs. This segment of classification pipeline classifies 3 primary types of Punches i.e., Uppercut, Hook, and Straight punch (Jab/Cross). For following part, 'other hand' refers to the inactive hand during the punch, which is mostly to get the threshold (assuming both hands are identical). In AlphaPose results, y coordinate is vertical pixel value and x coordinate is horizontal pixel values. All measured distances (l_{pixel}) are taken relative to the middle point of the shoulders, which remains relatively stable during the punch action. l_{pixel} are proportional to the real-world distances (l_{real}). It's valid to assume that all body parts are approximately at same depth from camera, hence we can define an unknown depth dependent proportionality constant α , such $l_{\text{real}} = \alpha * l_{\text{pixel}}$. All the rules compare l_{pixel} of various parts, α doesn't affect the decision. The distance represented by 'l' corresponds to 2D Euclidean distance, while 'x' or 'y' represents the difference between corresponding coordinates.

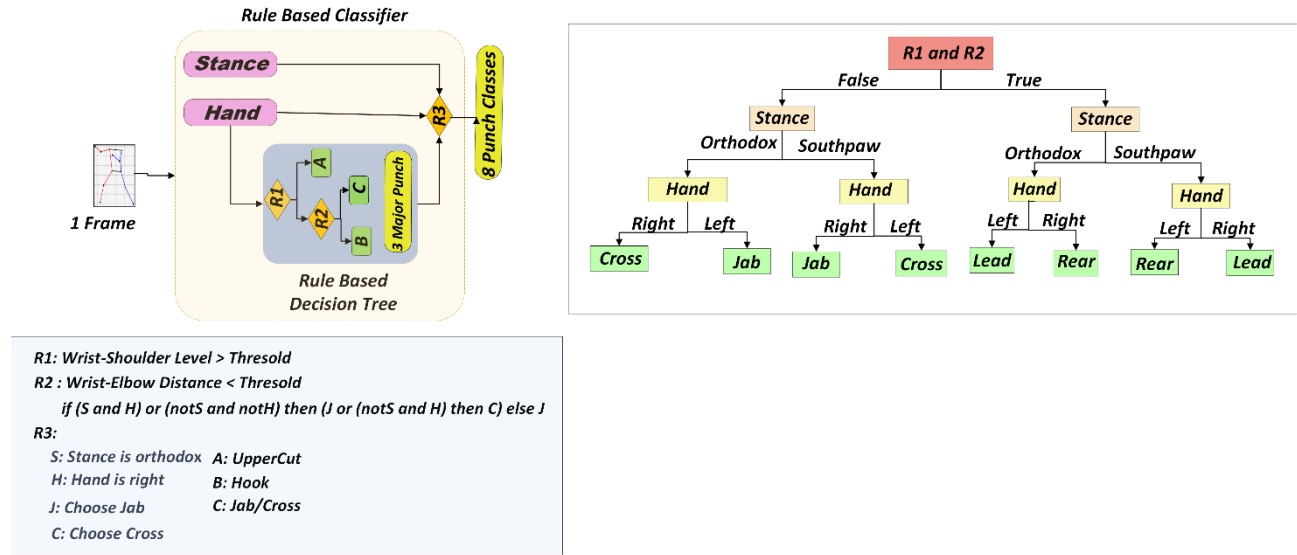


Figure 2. Rule-Based Classifier pipeline and R3 rule binary tree visualization

- **Rule 1 (Uppercut Detection):**

We scrutinize whether a punch qualifies as an uppercut by assessing the difference in wrist-shoulder heights (Δ_{wsy}). If this difference exceeds a predefined threshold of two-fifths of the other hand wrist-elbow height difference ($\frac{2}{5} \Delta_{wey'}$), the result is True, and we classify the punch as an uppercut. Else, R1 is False. Here the wrist elbow distance of the other hand is the same as the forearm length.

$$R1 = \{ True, \Delta_{wsy} > \frac{2}{5} \Delta_{wey'} \text{ False}, \Delta_{wsy} \leq \frac{2}{5} \Delta_{wey'} \} \quad (3)$$

- **Rule 2 (Hook Detection):**

To classify a punch as a hook, we analyze the side view of the punch action. Ideally, the wrist and elbow should occlude each other during a hook. Based on empirical findings, we ascertain this by verifying if the wrist-elbow distance (Δ_{wel}) is less than half of the corresponding measure for the other hand ($\Delta_{wel'}$). If the distance is less than the threshold, R2 is True and Classified as Hook. Else, R2 is False.

$$R2 = \{ True, \Delta wsl < \frac{1}{2} \Delta wsl' \quad False, \Delta wsl \leq \frac{1}{2} \Delta wsl' \} \quad (4)$$

- **Rule 3 (Jab, Cross, Rear, or Lead Classification):**

This combines information from above rules to classify the input into 7 classes .R3 is a simple set of if-else conditions that either decide whether a punch is Jab or Cross if it's not classified as uppercut or hook in the preceding rules, in case of previously classified as uppercut or hook then we decide whether the punch is a rear or lead version of the detected punches. The entire R3 rule determines whether a punch is a rear or lead type; for instance, a jab is considered a lead punch, while a cross is the rear version of the same punch. R3 can be visualized as a binary tree in Figure 2.

In the end, we have seven classes, i.e., No Punch, Jab, Cross, Lead Uppercut, Rear Uppercut, Lead Hook, and Rear Hook. All the thresholds used are empirical, based on observations of larger, unstructured boxing poses from open-source and publicly available videos, which kept the generality and robustness to real-world data. Hardcoded algorithms are susceptible to various conditions of input data. Nevertheless, we build this classifier using relative features, which minimizes any player-based performance variance. This method is best for high-speed actions, where segmenting video for a particular punch is difficult, where this method can be used for classifying each frame and then considering mode-based decisions over a 20 frames window to classify the whole sequence.

4.2. Machine Learning-Based Classification

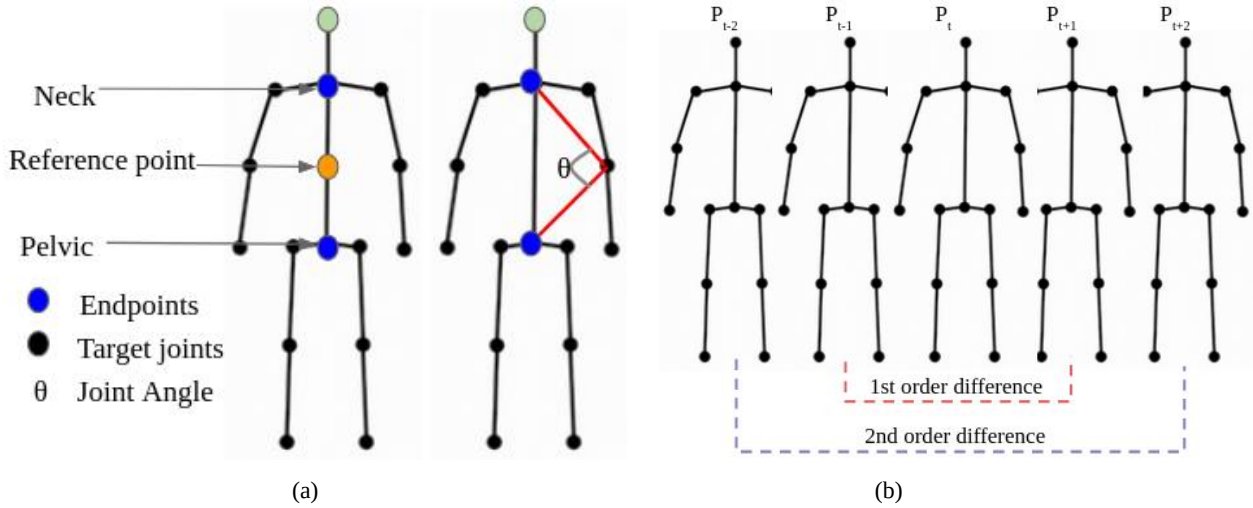


Figure 3. Feature Extraction (a) Unified-Axis angular encoding, (b) 2D Motion Dynamics Descriptors

4.2.1 Feature Extraction

In this study, our target is to correctly classify the fine-grained action i.e., boxing punch type, by using vision-only approach in 2D space. The complete dataset consists of 1334 pose sequences partitioned into 14 punch classes. Each pose sequence consists of 20 frames, and each frame consists of 12 key-body joint locations as 2D camera coordinates. To make the pose sequences translational and rotational invariant, absolute coordinates are converted into relative coordinates. For this purpose, in each frame, out of 12 key-points, the average of two most stable joints (i.e., neck and pelvic) is considered as a candidate for reference point. To encode translational features, the average of all candidate reference points is taken as the ultimate reference point for a particular pose sequence Eq. 5.

$$\sum_{n=0}^{N-1} \sum_{j=1, j \in \{s\}} s_j / 2N \quad (5)$$

For fine-grained classification, features should be highly sensitive to motion and generalized but robust to noise. To achieve this, we proposed following three human motion dynamics representations features:

1. Unified-axis Angular Encoding (UAE):

Inspired by the graph node embedding attributes for skeleton-based action recognition using Graph Neural Networks^[28], we propose a fourth-order feature to depict the relative movements between body parts. These

features measure the angles of highly moving joints with respect to the stable joints as shown in Fig. 3(a), and encode the pose sequence. Given three joints, $\ddot{u}$, s_1 , and s_2 , where $\ddot{u}$ target moving joints to calculate the angular features and s_1 and s_2 are the stable joints in the skeleton, $\mathbf{b}_{\ddot{u}s_i} = (\mathbf{x}_{s_i} - \mathbf{x}_{\ddot{u}}, \mathbf{y}_{s_i} - \mathbf{y}_{\ddot{u}})$ denotes the vector from joint $\ddot{u}$ to s_i ($i = 1, 2$), where (x_k, y_k) represent the 2D coordinates of joint k ($k = \ddot{u}, s_1, s_2$). Considering these angles, we define two angular features to encode the complete pose sequence:

a. Static Angular Encoding:

Static encoding defines joints' pose or relative location within a particular frame with the help of the angle between three joints. Despite first order and second-order features, representing the coordinates of a joint and the lengths and directions of bones respectively, these joint angles are third-order features. These features are inclined towards relative motion and are human subject scale invariant. Let's take θ as the angle between $\mathbf{b}_{\ddot{u}s_1}$ and $\mathbf{b}_{\ddot{u}s_2}$; the static angular encoding $d(\ddot{u})$ for joint $\ddot{u}$ is defined by the equation Eq. 6 as

$$d(\ddot{u}) = 1 - \cos \theta = 1 - \frac{\mathbf{b}_{\ddot{u}s_1} \cdot \mathbf{b}_{\ddot{u}s_2}}{\|\mathbf{b}_{\ddot{u}s_1}\| \|\mathbf{b}_{\ddot{u}s_2}\|} \quad (6)$$

b. Velocity Angular Encoding:

Velocity encoding defines the variation of the joints' angles or movement across the frames. It involves the temporal features determined by the forward difference of the static angular features. Since we are adding an extra dimensionality of time, these features become fourth order. Angular velocity v at frame (t) is given by the equation Eq. 7 as

$$v_t = d(\ddot{u}_t) - d(\ddot{u}_{t-1}) \quad (7)$$

2. Unified-axis Angular Encoding (UAE):

Since sports activities are highly dynamic and most of the sports-specific movements are accelerated motion, it is required to encode the velocity and acceleration of the poses (joint locations)^[29]. For this purpose, we propose a Taylor series based encoding scheme as shown in Fig. 3(b). Let's suppose that a pose sequence $\mathbf{P}$ is represented as a set of N number of poses i.e. $\{\mathbf{P}_0, \mathbf{P}_1, \dots, \mathbf{P}_{t-1}, \mathbf{P}_t, \mathbf{P}_{t+1}, \dots, \mathbf{P}_{N-2}, \mathbf{P}_{N-1}\}$. Velocity ($\mathbf{P}_v$) and acceleration ($\mathbf{P}_a$) are encoded as the first and second central differences at a particular instant of time. So, velocity at time t ($\mathbf{v}_t$) is given as $\mathbf{P}_{t+1} - \mathbf{P}_{t-1}$ whereas acceleration at time t ($\mathbf{a}_t$) is given as $\mathbf{P}_{t+2} + \mathbf{P}_{t-2} - 2\mathbf{P}_t$. The overall second-order Taylor series approximated feature f_{taylor} is given by the equation Eq. 8.

$$f_{taylor}^t = [\mathbf{P}_t; \mathbf{P}_{v_t}; \mathbf{P}_{a_t}] \quad (8)$$

3. Fifth-order Angular Encoding:

These features are hybrid of both unified axis angular encoding and 2D motion dynamics descriptors. For a pose sequence having N number of frames, J number of moving joints and two stable joints, joint angles $\theta_{t\ddot{u}}$ are calculated for each moving joint ($\ddot{u}_j$), $j = \{1, \dots, J\}$ to derive static angular encoding for each pose frame. Let's suppose θ_t be a joint angle vector having J elements at time t ($\theta_{t\ddot{u}_j}$), where $j = \{1, \dots, J\}$. Static angular encoding for a pose sequence θ can be represented as a set $\{\theta_0, \theta_1, \dots, \theta_{t-1}, \theta_t, \theta_{t+1}, \dots, \theta_{N-2}, \theta_{N-1}\}$. Applying Taylor series approximation on θ to encode velocity (θ_v) and acceleration (θ_a), we get the equation Eq. 9 as our fifth-order angular features $\mathbf{f}(\theta)$:

$$\mathbf{f}(\theta_t) = [\theta_t; \theta_{v_t}; \theta_{a_t}] \quad (9)$$

4.3. Machine Learning Classifiers

To test the generalizability of the hand-crafted motion features, we selected three classical machine learning algorithms for multi-class classification tasks^[30]. A complete end-to-end supervised training pipeline is shown in the Fig. 4.

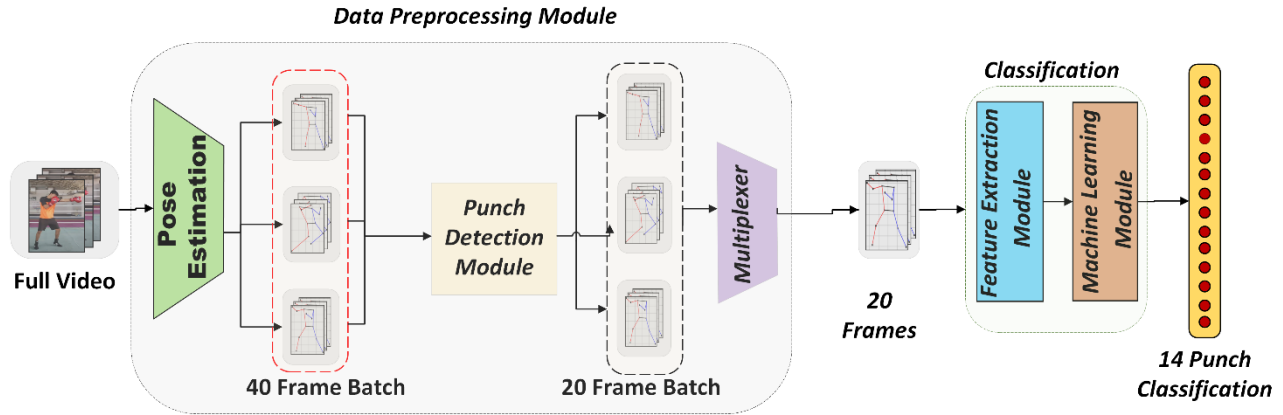


Figure 4. An End-to-End Machine Learning based pipeline for Fine-Grained Boxing Punch Classification

1. Support Vector Machine (SVM):

SVM is a supervised machine learning algorithm used for classification and regression tasks. It seeks to find the hyperplane that best separates data points into different classes while maximizing the margin between classes. SVM works by finding support vectors, which are the data points closest to the decision boundary (hyperplane). It can handle linear and non-linear data separations by using different kernel functions (e.g., linear, polynomial, radial basis function). In our case, we used $C = 10$, radial basis function for kernel, balanced class weight and one-vs-rest decision function approach.

2. Random Forest (RF):

Random Forest is an ensemble learning method that combines multiple decision trees to make predictions. It creates a forest of decision trees, each trained on a random subset of the data and features. RF aggregates the predictions of individual trees to improve accuracy and reduce overfitting. It provides estimates of feature importance based on the reduction in impurity (e.g., Gini impurity) achieved by each feature. In our case, we used 100 estimators, entropy function for impurity reduction, and balanced class weight.

3. K Nearest Neighbors (KNN):

k-Nearest Neighbors is a non-parametric supervised learning algorithm used for classification and regression. It classifies a data point based on the majority class among its k-nearest neighbors in the feature space. KNN can use various distance metrics (e.g., Euclidean, Manhattan) to measure similarity between data points. The choice of k (the number of neighbors) is a critical hyperparameter affecting the algorithm's performance. In our case, we used distance function to weight the data points and employed k-fold cross validation for hyperparameter tuning.

5. RESULTS AND DISCUSSION

5.1. Rule-Based Classification

In this section, we describe the experiments conducted to evaluate the Rule-Based Classifier. We tested this classifier using two methods: 1. Individual Frame and 2. Sequence of Frames. For individual labeled frame data, we applied the Rule-Based Classifier to both a controlled environment dataset and publicly available amateur boxing videos. In the case of amateur videos for Rule-Based Classifier, we manually trimmed off front and back view frames and generated a single-frame dataset using the Dataset Creation module.

The Rule-Based classification pipeline comprises four key components that influence our overall accuracy. Firstly, the punch detection component shines with an accuracy rate of 99.5% when distinguishing no punches from other actions. The small 0.5% misclassification occurs mainly during high-speed actions and self-occlusion instances. On the other hand, our hand and stance detection exhibit flawless performance, achieving a 100% accuracy rate. Within our rule-based algorithms, including R1, R2, and R3, R2 stands out as the least accurate, achieving only 54% accuracy. Nevertheless, it plays a crucial role in differentiating between Hook and jab/cross punches, even when facing challenges like self-occlusion

due to torso rotation and camera view deviations. Meanwhile, R1 and R3 both maintain accuracy rates above 90%. Overall, our system achieves an accuracy of 80.14% across seven classes.

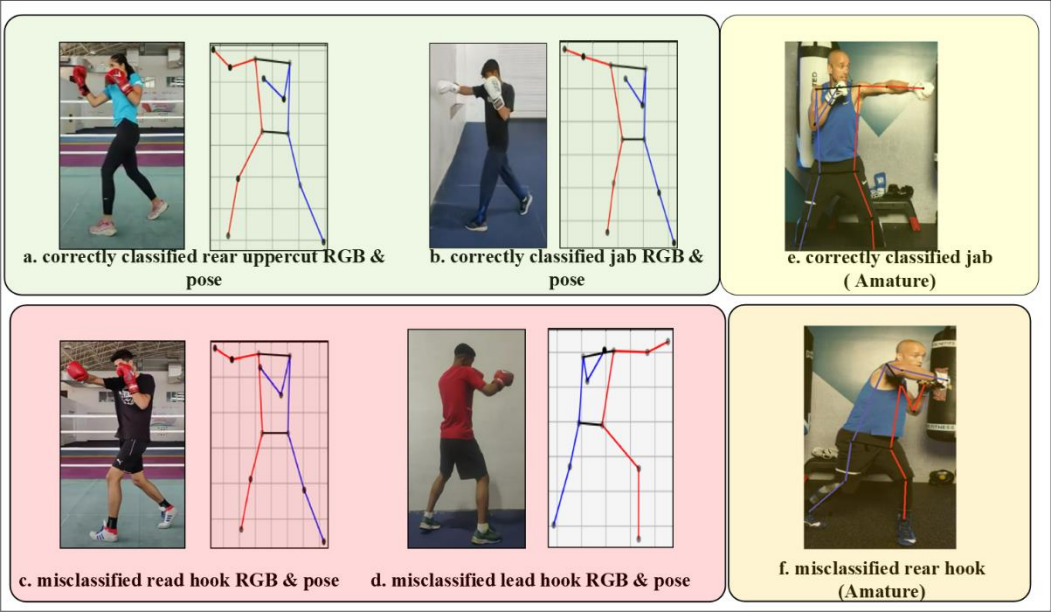


Figure 5. Poses corresponding to correct and misclassified results of Rule-Based Classifier on Controlled & Amateur Video

In the case of Sequence of Frames, as mentioned in Section 3, we classified each frame and used a mode-based decision over 20 frames. This method yielded a similar accuracy to individual punch classification, demonstrating its suitability for sequence classification as well. This level of performance remains consistent when classifying a diverse set of publicly available amateur boxing videos, where Jab, Cross, and Uppercut punches are identified with over 80% accuracy. In comparison, Hook punches exhibit a slightly lower accuracy of 47%. Instances of successful classification and misclassification for both controlled environment and wild-setting videos are illustrated in Figure 5. Figure 6 shows the confusion matrix of the rule-based classifiers for a mixed dataset consisting of controlled and wild-setting videos.

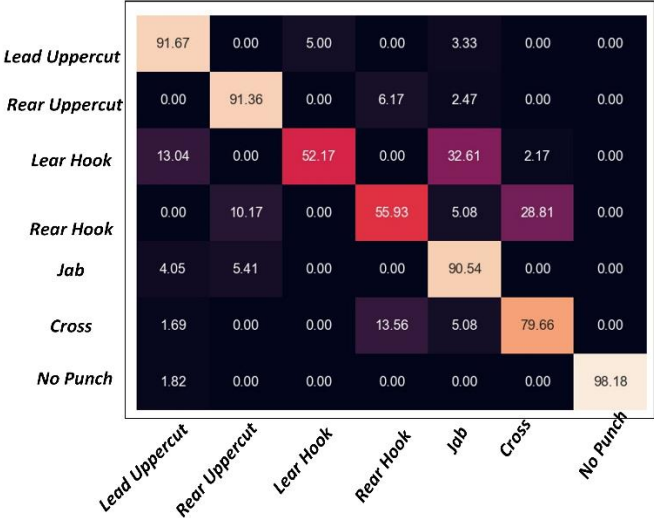


Figure 6. Rule-Based Classification Confusion Matrix for Mixed Dataset

5.2. ML-Based Classification

We derived three sets of features as mentioned above for the complete dataset and applied three classical machine learning classifiers for 14 type boxing punch classification tasks. For hyperparameter tuning and bias checking, 10-folds cross-validation is employed during the training while ensuring same seed for random shuffling. Accuracy, Precision, Recall and F1-Score are taken as the standard performance metrics. SVM, RFC and KNN classifiers are implemented for each set of features and trained for 14-level classification tasks. Table 3 reports the mean and standard deviation of each performance metric, taken over a 10-fold cross-validation dataset, are reported.

The optimal value of nearest neighbor parameter K in KNN is calculated using the brute-force search by varying the value of K from 1 to 20 and plot the accuracy vs K curve as shown in Fig. 7. Considering the sudden change in the slope of the curve, the optimum values of K for unified-axis angular encoding, 2D motion dynamics descriptors and fifth-order angular encoding are found to be 6, 3, 4 respectively. An overall confusion matrix over 10-fold cross-validation dataset for each combination of ML models and features are shown in Fig. 8. As per our literature review, it is the first work in fine-grained sports-specific skeleton-based action recognition work using vision-only, especially in boxing, so there is no direct benchmarking with the state-of-the-art. We will discuss only our results in the following paragraph.

Table 3. Information on video and audio files that can accompany a manuscript submission

ML Classifiers	Performance metrics	2D Motion Dynamic Descriptor	Unified-Axis Angular Encoding	Fifth-order angular encoding
Support Vector Machine	Accuracy	85 $\pm$ 1 %	33 $\pm$ 4 %	69 $\pm$ 4 %
	Precision	88 $\pm$ 2 %	36 $\pm$ 4 %	72 $\pm$ 4 %
	Recall	85 $\pm$ 1 %	33 $\pm$ 4 %	69 $\pm$ 4 %
	F1 Score	85 $\pm$ 1 %	33 $\pm$ 4 %	69 $\pm$ 4 %
Random Forest	Accuracy	92 $\pm$ 3 %	37 $\pm$ 3 %	85 $\pm$ 3 %
	Precision	93 $\pm$ 2 %	43 $\pm$ 7 %	86 $\pm$ 2 %
	Recall	92 $\pm$ 3 %	37 $\pm$ 3 %	85 $\pm$ 3 %
	F1 Score	92 $\pm$ 3 %	36 $\pm$ 3 %	85 $\pm$ 3 %
K Nearest Neighbor	Accuracy	98 $\pm$ 6 %	91 $\pm$ 25 %	99 $\pm$ 4 %
	Precision	98 $\pm$ 6 %	92 $\pm$ 22 %	99 $\pm$ 4 %
	Recall	98 $\pm$ 6 %	91 $\pm$ 25 %	99 $\pm$ 4 %
	F1 Score	98 $\pm$ 6 %	91 $\pm$ 25 %	98 $\pm$ 4 %

As we can apparently observe from the Table 3 that the average accuracies of three different classical ML models, using Taylor series driven features having velocity and acceleration encoding, are 85%, 92% and 98% for SVM, RFC and KNN respectively, which are much higher and stable. However, unified-axis angular encoding-based features are reported with 33% and 37% accuracy with stability in the case of SVM and RFC respectively. Although, it's showing higher accuracy of 91% with KNN model but the results are highly unstable with 25% standard deviation. It has been found that if the hybrid features are employed to encode the velocity and acceleration in the joint angles itself using Taylor series approximation, we can enhance the accuracy of the classifiers along with the stability in the results. It's evident from the table that the average accuracies of the models with fifth-order angular features have been increased by approximately 2 times, i.e., 69%, 85% and 99% with SVM, RFC and KNN respectively.

We have shown the implementation of a lightweight ML pipeline for fine-grained sports-specific action recognition i.e., skeleton-based boxing punches type classification in our case, for metaverse sports applications.

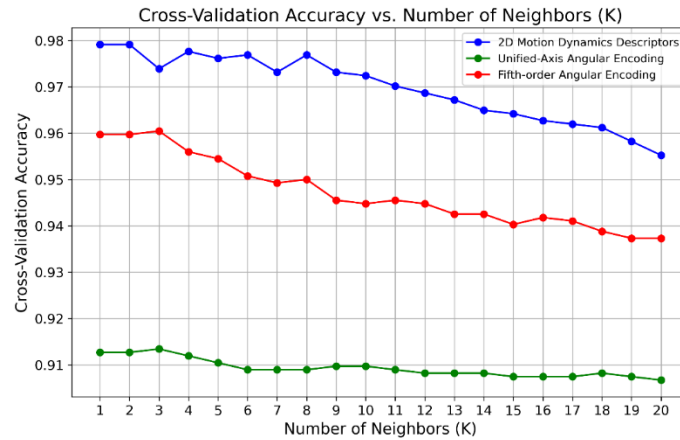


Figure 7. Hyperparameter tuning of K-Nearest Neighbors using 10-fold cross-validation

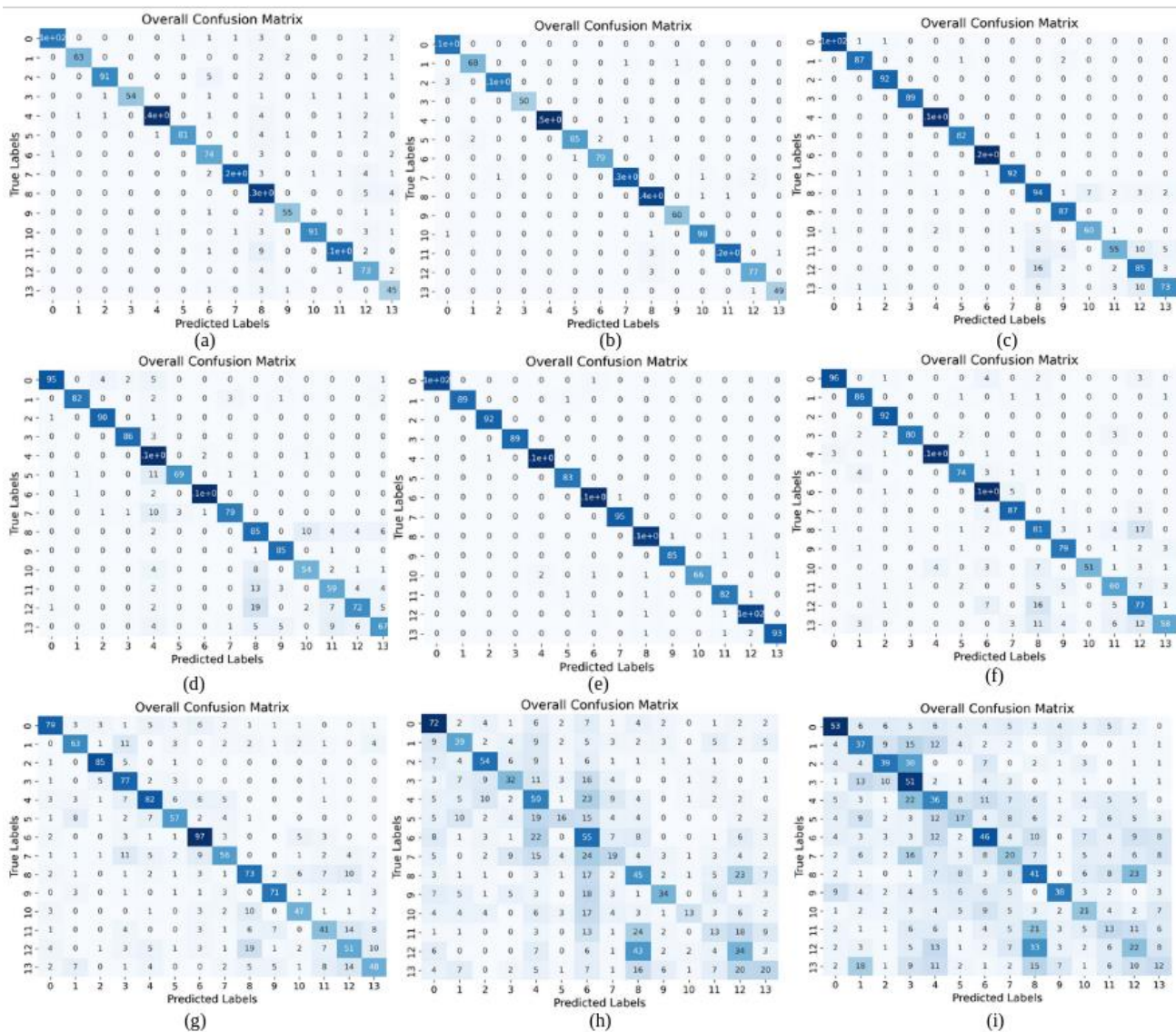


Figure 8. Confusion matrix for the 14-types boxing punch classification using (a) SVM with FAE, (b) RFC with FAE, (c) KNN with FAE, (d) SVM with 2DMDD (e) RFC with 2DMDD (f) KNN with 2DMDD (g) SVM with UAE, (h) RFC with UAE, and (i) KNN with UAE

6. CONCLUSION

In this study, we have presented a novel approach for real-time action recognition in combat sports, specifically focusing on boxing, utilizing RGB video data. Our proposed pipeline incorporates three innovative motion dynamics features, namely Unified-Axis Angular Encoding (UAE), 2D Motion Dynamics Descriptors (2DMDD), and Fifth-order Angular Encoding (FAE), to capture and encode the velocity and acceleration of pose sequences. We employed classical machine learning algorithms, including Support Vector Machine (SVM), Random Forest (RF), and K Nearest Neighbors (KNN), to create a lightweight model for punch type classification. Our experimental results have demonstrated promising performance in classifying 14 different boxing punch types using these features. The average accuracies achieved are 55% for UAE, 92% for 2DMDD, and 84% for FAE, respectively. Particularly noteworthy is the remarkable 99% accuracy achieved on 10-fold cross-validation using the FAE features with the KNN algorithm, showcasing the potential of our approach for accurate and real-time classification of boxing punches.

REFERENCES

- [1] Anthes, C., García-Hernández, R. J., Wiedemann, M., & Kranzlmüller, D. (2016, March). State of the art of virtual reality technology. In *2016 IEEE aerospace conference* (pp. 1-19). IEEE.
- [2] Wohlgenannt, I., Simons, A., & Stieglitz, S. (2020). Virtual reality. *Business & Information Systems Engineering*, 62, 455-461.
- [3] dos S Silva, F. H., Bezerra, G. M., Holanda, G. B., de Souza, J. W. M., Rego, P. A., Neto, A. V. L., ... & Rebouças Filho, P. P. (2021). A novel feature extractor for human action recognition in visual question answering. *Pattern Recognition Letters*, 147, 41-47.
- [4] Kasiri, S., Fookes, C., Sridharan, S., & Morgan, S. (2017). Fine-grained action recognition of boxing punches from depth imagery. *Computer Vision and Image Understanding*, 159, 143-153.
- [5] Yi, Y., Li, A., & Zhou, X. (2020). Human action recognition based on action relevance weighted encoding. *Signal Processing: Image Communication*, 80, 115640.
- [6] Mohith, S. S., Vijay, S., Sanjana, V., & Krupa, N. (2020, December). Trajectory based human action recognition using centre symmetric local binary pattern descriptors. In *2020 IEEE 17th India Council International Conference (INDICON)* (pp. 1-6). IEEE.
- [7] Abdul-Azim, H. A., & Hemayed, E. E. (2015). Human action recognition using trajectory-based representation. *Egyptian Informatics Journal*, 16(2), 187-198.
- [8] Papadopoulos, K., Antunes, M., Aouada, D., & Ottersten, B. (2017, September). Enhanced trajectory-based action recognition using human pose. In *2017 IEEE international conference on image processing (ICIP)* (pp. 1807-1811). IEEE.
- [9] Simonyan, K., & Zisserman, A. (2014). Two-stream convolutional networks for action recognition in videos. *Advances in neural information processing systems*, 27.
- [10] Wang, L., Xiong, Y., Wang, Z., Qiao, Y., Lin, D., Tang, X., & Van Gool, L. (2016). Temporal segment networks: Towards good practices for deep action recognition. In *European conference on computer vision* (pp. 20-36). Springer, Cham.
- [11] Sigurdsson, G. A., Divvala, S., Farhadi, A., & Gupta, A. (2017). Asynchronous temporal fields for action recognition. In *Proceedings of the IEEE conference on Computer Vision and Pattern Recognition* (pp. 585-594).
- [12] Shah, A., Mishra, S., Bansal, A., Chen, J. C., Chellappa, R., & Shrivastava, A. (2022). Pose and joint-aware action recognition. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision* (pp. 3850-3860).
- [13] Jhuang, H., Gall, J., Zuffi, S., Schmid, C., & Black, M. J. (2013). Towards understanding action recognition. In *Proceedings of the IEEE international conference on computer vision* (pp. 3192-3199).
- [14] Luo, Y., Ren, J., Wang, Z., Sun, W., Pan, J., Liu, J., ... & Lin, L. (2018). Lstm pose machines. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 5207-5215).
- [15] Amerineni, R., Gupta, L., Steadman, N., Annauth, K., Burr, C., Wilson, S., ... & Vaidyanathan, R. (2021). Fusion models for generalized classification of multi-axial human movement: Validation in sport performance. *Sensors*, 21(24), 8409.

- [16] Zhao, B., Lu, H., Chen, S., Liu, J., & Wu, D. (2017). Convolutional neural networks for time series classification. *Journal of Systems Engineering and Electronics*, 28(1), 162-169.
- [17] Hatami, N., Gavet, Y., & Debayle, J. (2018, April). Classification of time-series images using deep convolutional neural networks. In *Tenth international conference on machine vision (ICMV 2017)* (Vol. 10696, pp. 242-249). SPIE.
- [18] Sigal, L., Balan, A. O., & Black, M. J. (2010). Humaneva: Synchronized video and motion capture dataset and baseline algorithm for evaluation of articulated human motion. *International journal of computer vision*, 87(1-2), 4-27.
- [19] A. Stisen et al., "Smart devices are different: Assessing and mitigating mobile sensing heterogeneities for activity recognition," in *Proc. ACM Conf. Embedded Netw. Sensor Syst.*, 2015, pp. 127–140.
- [20] Kong, Q., Wu, Z., Deng, Z., Klinkigt, M., Tong, B., & Murakami, T. (2019). Mmact: A large-scale dataset for cross modal human action understanding. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (pp. 8658-8667).
- [21] Fang, H. S., Li, J., Tang, H., Xu, C., Zhu, H., Xiu, Y., ... & Lu, C. (2022). Alphapose: Whole-body regional multi-person pose estimation and tracking in real-time. *IEEE Transactions on Pattern Analysis and Machine Intelligence*.
- [22] Cao, Z., Simon, T., Wei, S. E., & Sheikh, Y. (2017). Realtime multi-person 2d pose estimation using part affinity fields. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 7291-7299).
- [23] Toshev, A., & Szegedy, C. (2014). Deeppose: Human pose estimation via deep neural networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 1653-1660).
- [24] Zheng, C., Wu, W., Chen, C., Yang, T., Zhu, S., Shen, J., ... & Shah, M. (2020). Deep learning-based human pose estimation: A survey. *ACM Computing Surveys*.
- [25] Zhang, F., Zhu, X., & Wang, C. (2022). A Comprehensive Survey on Single-Person Pose Estimation in Social Robotics. *International Journal of Social Robotics*, 14(9), 1995-2008.
- [26] Kharb, U., & Gautam, T. (2022). Review and Analysis of Various Human Pose Estimation Models. *Available at SSRN 4157643*.
- [27] Kimm, D., & Thiel, D. V. (2015). Hand speed measurements in boxing. *Procedia Engineering*, 112, 502-506.
- [28] Qin, Z., Liu, Y., Ji, P., Kim, D., Wang, L., McKay, R. I., ... & Gedeon, T. (2022). Fusing higher-order features in graph neural networks for skeleton-based action recognition. *IEEE Transactions on Neural Networks and Learning Systems*.
- [29] M. Zanfir, M. Leordeanu, C. Sminchisescu, The moving pose: an efficient 3d kinematics descriptor for low-latency action recognition and detection, in: *Proceedings of International Conference on Computer Vision (ICCV)*, IEEE, Sydney, Australia, 2013, pp. 2752–2759.
- [30] dos S Silva, F. H., Bezerra, G. M., Holanda, G. B., de Souza, J. W. M., Rego, P. A., Neto, A. V. L., ... & Rebouças Filho, P. P. (2021). A novel feature extractor for human action recognition in visual question answering. *Pattern Recognition Letters*, 147, 41-47.